Network Security and Cryptography

2nd September 2024

Marking Scheme

Markers are advised that, unless a task specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) **must** be given the available marks. If there is doubt as to the correctness of an answer, the relevant NCC Education materials should be the first authority.

This marking scheme has been prepared as a **guide only** to markers and there will frequently be many alternative responses which will provide a valid answer.

Each candidate's script must be fully annotated with the marker's comments (where applicable) and the marks allocated for each part of the tasks.

Throughout the marking, please credit any valid alternative point.

Where markers award half marks in any part of a task, they should ensure that the total mark recorded for the task is rounded up to a whole mark.

Marker's comments:

Moderator's comments:

Mark:	**Moderated mark:**	**Final mark:**

Penalties applied for academic malpractice:

Answer ALL questions

Marks

Question 1

a) Explain the **THREE (3)** fundamental principles of network security. **6**

Mark Scheme

Award 2 marks for each principle up to a maximum of 6 marks:

- ***Confidentiality (1 mark) refers to the protection of sensitive data from unauthorised access or disclosure. This principle ensures that only authorised individuals or systems can access and view confidential information (1 mark).***
- ***Integrity (1 mark) ensures that data remains accurate, complete, and unaltered by unauthorised parties during transmission or storage (1 mark).***
- ***Availability (1 mark) ensures that authorised users and systems have reliable and timely access to information and resources when needed (1 mark).***

b) Alice wants to send a message to Bob using symmetric encryption. Describe what a **Message Authentication Code (MAC)** is and how Alice can use it to ensure the data integrity of the message received by Bob. **4**

Mark Scheme

Award 2 marks for each of the following up to a maximum of 4 marks:

- ***A Message Authentication Code (MAC) is a cryptographic code or tag derived from the message data and a shared secret key between Alice and Bob, which provides data integrity and authentication (2 marks).***

- ***This MAC is transmitted alongside the message, allowing Bob to verify the integrity and authenticity of the message by decrypting the message and verifying the MAC to ensure the encrypted data's integrity (2 marks).***

Total 10 Marks

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Marks

Question 2

a) Describe **TWO (2)** key differences between Block **and** Stream Ciphers. **4**

Mark Scheme

Award 1 mark for each of the following up to a maximum of 4 marks:

- ***Block ciphers encrypt data in fixed-size blocks, typically 64 or 128 bits at a time. (1 mark)***
- ***Block ciphers require padding if the data to be encrypted is not an exact multiple of the block size. (1 mark)***
- ***Stream ciphers encrypt data one bit or byte at a time. (1 mark).***
- ***Stream ciphers do not require padding, as they can encrypt data of any length. (1 mark).***

b) Describe **THREE (3)** methods utilised for sharing a secret key between two parties in symmetric cryptography. **6**

Mark Scheme

Award 2 marks for any of the following up to a maximum of 6 marks:

- ***Pre-Shared Key (PSK) (1 mark): Both parties agree on a secret key in advance before communication begins. (1 mark)***
- ***Key Distribution Centre (KDC) (1 mark): A trusted third party facilitates the secure distribution of the secret key to both parties. (1 mark)***
- ***Diffie-Hellman Key Exchange (1 mark): Enables two parties to establish a shared secret key over an insecure channel without transmitting the key directly. (1 mark)***
- ***Public Key Infrastructure (PKI) (1 mark): Uses public and private key pairs to securely distribute symmetric keys for encryption. (1 mark)***

Total 10 Marks

Question 3

a) Explain the difference between Biometric Authentication **and** Multi-Factor Authentication (MFA). **4**

Mark Scheme

Award 2 marks for each of the following up to a maximum of 4 marks:

- ***Biometric authentication relies on unique physical or behavioural characteristics of individuals, such as fingerprints, iris patterns, or facial recognition (2 marks).***

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Marks

- ***MFA combines two or more independent authentication factors, such as something you know (password), something you have (token or OTP) and something you are (biometric) (2 marks)***

b) Compare and contrast discretionary access control (DAC), mandatory access control (MAC) and role-based access control (RBAC). **6**

Mark Scheme

Award 2 marks for each of the following up to a maximum of 6 marks:

- ***Discretionary Access Control (DAC): Users have control over the access permissions of resources they own, allowing them to grant or revoke access at their discretion. (2 marks)***
- ***Mandatory Access Control (MAC): Access permissions are centrally controlled by system administrators or security policies, based on labels or security classifications assigned to resources and users. (2 marks)***
- ***Role-Based Access Control (RBAC): Access permissions are based on predefined roles or job functions assigned to users, regulating their access to resources based on their roles rather than individual permissions. (2 marks)***

Total 10 Marks

Question 4

a) State what password salting is **and** why is it crucial for password security. **4**

Mark Scheme

Award 1 mark for each of the following up to a maximum of 4 marks:

- ***Password salting involves adding a random sequence of characters to a password before hashing it. (1 mark)***
- ***Prevents attackers from using precomputed rainbow tables. (1 mark)***
- ***Mitigates Dictionary Attacks. (1 mark)***
- ***Salting adds an additional layer of complexity to the hashing process. It ensures that identical passwords result in different hashed values. (1 mark)***

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Marks

b) Explain what input validation is **and** why it is essential for preventing security vulnerabilities. **6**

Mark Scheme

Award 2 marks each for the following up to a maximum of 6 marks:

- ***Input validation is the process of inspecting and validating data entered by users to ensure it meets specified criteria before processing it. (2 marks)***
- ***By implementing input validation, developers can verify that input does not contain malicious commands such as SQL injection (2 marks)***
- ***Input validation helps detect and block input containing script tags or other potentially harmful content (2 marks)***

Total 10 Marks

Question 5

a) Discuss **THREE (3)** common characteristics of weak passwords and their security implications. **6**

Mark scheme

Award 2 marks for any of the following up to a maximum of 6 marks:

- ***Weak passwords are often short in length, typically consisting of fewer than eight characters (1 mark). Such passwords are more susceptible to brute-force attacks. (1 mark)***
- ***Weak passwords frequently lack complexity (1 mark), relying on simple patterns, dictionary words or easily guessable sequences. (1 mark)***
- ***Weak passwords are often reused across multiple accounts (1 mark) or systems, lacking diversity or uniqueness. (1 mark)***

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Marks

b) Explain **TWO (2)** key differences between a hash function **and** a one-time password (OTP). **4**

Mark scheme

Award 1 mark for each of the following up to a maximum of 4 marks:

- ***A hash function is a mathematical algorithm that takes an input (or message) and produces a fixed-size output called a hash value (or digest). (1 mark)***
- ***The primary purpose of a hash function is data integrity and verification. (1 mark)***
- ***A one-time password (OTP) is a temporary code or password that is valid for only one authentication session or transaction. (1 mark)***
- ***OTPs are used as an additional factor of authentication to enhance security beyond just a username and static password. (1mark)***

Total 10 Marks

Question 6

a) Explain what a buffer overflow is **and** how it can lead to significant software security vulnerabilities. **4**

Mark scheme

Award 1 mark for each of the following up to a maximum of 4 marks:

- ***A buffer overflow occurs when a program writes data beyond the bounds of a fixed-size buffer in memory. (1 mark)***
- ***Unauthorised Access: They can allow attackers to gain unauthorised access to systems by overwriting critical control information used by the system. (1 mark)***
- ***Code Execution: Attackers can exploit buffer overflows to execute arbitrary code, potentially taking control of the system to perform malicious actions. (1 mark)***
- ***Data Corruption: Buffer overflows can corrupt existing data, leading to unpredictable behaviour, crashes and system instability. (1 mark)***

b) Discuss **THREE (3)** recommendations you would propose to safeguard your computer system against malware. **6**

Mark Scheme

Award 2 marks each for any of the following up to a maximum of 6 marks:

- ***Install and regularly update reputable antivirus and antimalware software to detect and remove malicious software. (2 marks)***

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Marks

- ***Practice safe browsing habits, such as avoiding suspicious websites, not clicking on suspicious links or email attachments and being cautious when downloading files from untrusted sources. (2 marks)***
- ***Keep all software, including the operating system and applications up to date with the latest security patches and updates. (2 marks)***

Total 10 Marks

Question 7

a) Describe **THREE (3)** strategies that can be implemented to prevent a ransomware attack from encrypting all data on a university network. **6**

Mark scheme

Award 2 marks for each of the following up to a maximum of 6 marks:

- ***Regularly backup data and maintain offline copies to ensure data can be restored in case of a ransomware attack. (2 marks)***
- ***Implement robust network security measures, such as firewalls, intrusion detection systems and antivirus software, to detect and prevent ransomware infections. (2 marks)***
- ***Educate staff and students about phishing awareness and safe computing practices to reduce the likelihood of ransomware infiltration through malicious emails or websites. (2 marks)***

b) Explain how DKIM (Domain Keys Identified Mail) contributes to email security. **4**

Mark scheme

Award 1 mark for each of the following up to a maximum of 4 marks:

- ***DKIM is an email authentication method (1 mark) that uses digital signatures to verify that an email message was sent and authorised by the owner of a domain. (1 mark)***
- ***It adds a signature to the email header (1 mark), allowing the recipient to validate the sender's authenticity. (1 mark)***

Total 10 Marks

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Question 8

Marks

a) Explain what a firewall is **and** the difference between a hardware and software firewall. **6**

Mark scheme

Award 2 marks for each of the following up to a maximum of 6 marks:

- ***A firewall is a network security device or software that monitors and controls incoming and outgoing network traffic based on predetermined security rules. Its primary function is to act as a barrier between a trusted internal network and untrusted external networks. (2 marks)***
- ***A hardware firewall is a physical device that is typically installed at the boundary between a private network and the internet. (2 marks)***
- ***A software firewall is a program or application installed on a computer or server to control inbound and outbound network traffic. (2 marks)***

b) Discuss **ONE (1)** advantage and disadvantage of application-level proxy compared with packet filtering firewall. **4**

Mark scheme

Award 1 mark for each of the following up to a maximum of 4 marks:

- ***Enhanced Security (1 mark): One advantage of an application-level proxy is its ability to provide advanced security features, such as deep packet inspection, content filtering and application-layer awareness. (1 mark)***
- ***Performance Overhead (1 mark): One disadvantage of application-level proxies is the potential introduction of latency and overhead to network traffic. (1 mark)***

Total 10 Marks

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Marks

Question 9

a) Explain what a Virtual Private Network (VPN) is **and** the **TWO (2)** primary types of VPNs. **6**

Mark scheme

Award 2 marks for each of the following up to a maximum of 6 marks:

- ***VPN is a technology that creates a secure encrypted connection (1 mark) over an insecure public network, typically the Internet. (1 mark) Remote Access VPNs (1 mark) are designed to provide secure access to a private network for individual users or devices over the internet. (1 mark)***
- ***Site-to-site VPNs, (1 mark) also known as gateway-to-gateway VPNs, are used to establish secure connections between two or more geographically dispersed networks or sites. (1 mark))***

b) Identify and explain **TWO (2)** potential risks associated with using public Wi-Fi networks to connect to a corporate network. **4**

Mark scheme

Award 1 marks for each of the following up to a maximum of 4 marks:

- ***Man-in-the-Middle (MitM) Attacks (1 mark): Attackers can intercept and eavesdrop on the data transmissions between a user's device and the corporate network. (1 mark)***
- ***Malware and Unauthorised Access (1 mark): Public Wi-Fi networks are often unsecured or poorly secured, making it easier for attackers to distribute malware or gain unauthorized access to connected devices. (1 mark)***

Total 10 Marks

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Question 10

a) Explain the difference between a host-based intrusion detection system (HIDS) and a network-based intrusion detection system (NIDS) **and** where each one would be more suitable on a network. **4**

Mark scheme

Award 1 marks for each of the following up to a maximum of 4 marks:

- ***HIDS monitors activities on a single host (1 mark), while NIDS monitors network traffic for suspicious behaviour (1 mark).***
- ***In a scenario where critical servers need individual protection, a HIDS is more suitable (1 mark). For overall network security and detecting threats across multiple hosts, a NIDS is preferable. (1 mark)***

b) Explain risk assessment **and** the difference between qualitative and quantitative risk assessments in information security. **6**

Mark scheme

Award 1 marks for each of the following up to a maximum of 6 marks:

- ***Risk assessment is a crucial process in information security that involves identifying, analysing and evaluating potential risks to an organisation's information assets, systems and processes (1 mark).***
- ***It helps organisations understand the likelihood and impact of potential threats, (1 mark) enabling them to prioritise and implement appropriate risk mitigation strategies (1 mark).***
- ***Qualitative Risk Assessment :Involves subjective evaluation of risks, often using descriptive terms like low, medium or high(1 mark). It provides a broad understanding of risks but lacks precision (1 mark).***
- ***Quantitative Risk Assessment: Involves numerical analysis, assigning monetary values to risks (1 mark). It provides a more precise evaluation but requires detailed data (1 mark).***

Total 10 Marks

End of paper

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Learning Outcomes matrix

Question	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1	1,2	Yes
2	1,2	Yes
3	4	Yes
4	3,4	Yes
5	2,3,4	Yes
6	3,4	Yes
7	3,4	Yes
8	4	Yes
9	4,5	Yes
10	4,6	Yes

Grade descriptors

Learning Outcome	Pass	Merit	Distinction
Understand the basic concept and application of Network Security and Cryptography	Demonstrate adequate understanding and application of Network Security and Cryptography	Demonstrate robust understanding and application of Network Security and Cryptography algorithm	Demonstrate a highly comprehensive understanding and application of Network Security and Cryptography.
Critically evaluate the suitability of different Cryptographic Algorithms	Demonstrates a basic evaluation of different cryptographic algorithms, identifying some strengths and weaknesses.	Demonstrates a thorough evaluation of different cryptographic algorithms, with a detailed analysis of their strengths and weaknesses, and appropriate comparisons.	Demonstrates a comprehensive and critical evaluation of different cryptographic algorithms, with in-depth analysis, insightful comparisons, and a well-argued justification of their suitability in various contexts.
Understand the vulnerabilities and systematically evaluate the security risks in networks	Demonstrates a basic understanding of vulnerabilities and can evaluate security risks in networks with guidance.	Demonstrates a good understanding of vulnerabilities and can systematically evaluate security risks in networks with some analysis.	Demonstrates a comprehensive understanding of vulnerabilities and can critically and systematically evaluate security risks in networks with detailed analysis.
Critically analyse appropriate tools	Demonstrates a basic ability to	Demonstrates a good ability to	Demonstrates an excellent ability to

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and techniques for network security	analyse appropriate tools and techniques for network security, with limited critical evaluation.	analyse appropriate tools and techniques for network security, with a clear and reasoned critical evaluation.	critically analyse appropriate tools and techniques for network security, providing comprehensive and insightful evaluation.
Understand the wireless network security and associated protocols	Demonstrates a basic understanding of wireless network security and its associated protocols.	Demonstrates a good understanding of wireless network security and its associated protocols, with clear explanations and some examples.	Demonstrates a comprehensive understanding of wireless network security and its associated protocols, providing detailed explanations, examples, and critical insights.
Explain the concepts of Information Security and explain good practices in achieving security	Demonstrates a basic understanding of the concepts of Information Security and provides examples of good practice in achieving security.	Demonstrates a clear and detailed understanding of the concepts of Information Security, explaining them accurately and providing well-chosen examples of good practices in achieving security.	Demonstrates a comprehensive and insightful understanding of the concepts of Information Security, explaining them thoroughly and providing detailed, relevant examples of best practices in achieving security.

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